FKF-YPR Sensör Modellemesinin Yeniden Düşünülmesi: Destek Vektör Regresyonu ile Yüksek Verimli Hapsedilme Kaybı Tahmini
Rethinking PCF-SPR Sensor Modeling: High-Efficiency Confinement Loss Prediction via Support Vector Regression

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***Özetçe* —Fotonik Kristal Fiber tabanlı Yüzey Plazmon Rezonans (FKF-YPR) sensörlerinin tasarımı ve optimizasyonu, yüksek doğrukuklu ancak hesaplama açısından maliyetli olan tam vektörel sonlu elemanlar yöntemi (TV-SEY) simülasyonlarına dayanmaktadır. Literatürde son zamanlarda, simülatör darboğazlarını aşmak için Üretken Çekişmeli Ağlar (GAN'lar) ile eşleştirilmiş Derin Sinir Ağlarının (DNN'ler) kullanılması önerilmiştir. Bu derin öğrenme boru hatları hesaplama açısından ağır ve veri açlığı çekmektedir. Bu çalışmada, FKF-YPR hapsedilme kaybını tahmin etmek için optimize edilmiş, hafif bir Destek Vektör Regresyonu (DVR) modeli öneriyoruz. Sabit bir 7-1-1 konfigürasyon bölümü kullanarak, optimize edilmiş DVR ve GPR modellerimiz sırasıyla 0,121 ve 0,185 Ortalama Kare Hata (MSE) değerlerine ulaşmıştır. Bu sonuçlar, literatürde rapor edilen GAN-artırılmış sinir ağı (MSE 0,314) sonuçlarından önemli ölçüde daha iyidir. Son olarak, SHAP değerleri aracılığıyla dış kaplama parametrelerinin hapsedilme kaybı üzerindeki etkilerini belirliyor ve modeli 27,63 nm/RIU ortalama spektral duyarlılık tahminiyle doğruluyoruz.**

Anahtar Kelimeler—Fotonik kristal fiber; yüzey plazmon rezonans; destek vektör regresyonu; açıklanabilir yapay zeka; Bayesyen optimizasyon.

***Abstract*—The design and optimization of Photonic Crystal Fiber based Surface Plasmon Resonance (PCF-SPR) sensors heavily rely on full vectorial finite element method (FV-FEM) simulations, which are highly accurate but computationally expensive. Recent literature has proposed using Deep Neural Networks (DNNs) paired with Generative Adversarial Networks (GANs) to bypass simulator bottlenecks. While effective, these deep learning pipelines are computationally heavy and data-hungry. In this paper, we propose a highly optimized, lightweight Support Vector Regression (SVR) model to predict PCF-SPR confinement loss. Using a static 7-1-1 configuration split, our optimized SVR and GPR models achieved Mean Squared Errors (MSE) of 0.121 and 0.185 respectively. These results significantly outperform the state-of-the-art GAN-augmented neural network (MSE 0.314) reported in literature. Finally, we apply SHAP (SHapley Additive exPlanations) to identify outer cladding parameters as primary confinement drivers, and successfully validate the model by predicting an average spectral sensitivity of 27.63 nm/RIU.**

Keywords—Photonic crystal fiber; surface plasmon resonance; support vector regression; explainable AI; Bayesian optimization.

I. Introduction

Photonic Crystal Fibers (PCFs) are at the forefront of modern optical sensing due to their flexible structural design and remarkable light-guiding capabilities. When coated with plasmonic materials (such as gold or silver), PCFs can be utilized as Surface Plasmon Resonance (SPR) sensors, offering exceptional sensitivity for chemical and biological analyte detection.

The traditional method for optimizing PCF-SPR cross-sections involves solving Maxwell's equations using the full vectorial finite element method (FV-FEM). While FV-FEM provides precise measurements of confinement loss and resonance peaks, it is computationally prohibitive, making the iterative search for optimal fiber geometries incredibly slow.

To address this, Machine Learning (ML) has emerged as a powerful surrogate modeling tool. Chugh et al. [2] pioneered the use of Artificial Neural Networks (ANNs) to compute optical properties of PCFs, demonstrating significant speedups over numerical simulators. However, deep learning models are notoriously data-hungry. Zelaci et al. [1] addressed this by introducing Generative Adversarial Networks (WGAN-GP) to synthesize thousands of artificial training samples, driving the predictive Mean Squared Error (MSE) down to 0.314. While effective, this GAN-ANN pipeline adds complexity and potential synthetic bias.

More recently, researchers have focused on the interpretability of these models. Khatun and Islam [3] utilized

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explainable AI (XAI) via SHAP values to optimize high-sensitivity biosensors, identifying critical design parameters such as gold thickness and analyte refractive index. Other traditional techniques, such as k-nearest neighbor regression, have also been explored for bent fiber modeling [4].

In this work, we hypothesize that the problem space of PCF-SPR confinement loss is better suited for mathematically rigorous, boundary-focused traditional algorithms that excel in small-data regimes. We evaluate Bayesian-optimized Support Vector Regression (SVR) and Gaussian Process Regression (GPR) against the GAN-ANN baseline. We demonstrate that these models can outperform the literature's augmented benchmarks without requiring any synthetic data generation. Finally, we apply SHAP analysis to extract physical design insights, providing a bridge between high-accuracy regression and engineering intuition.

II. Numerical Modeling and Dataset Generation

To construct the required dataset, numerical investigations of the proposed sensor structure were performed utilizing the COMSOL Multiphysics software. Utilizing Perfectly Matched Layers (PMLs), the Full Vectorial Finite Element Method (FV-FEM) was the primary tool employed for these simulations. Illustrated in Fig. 1 is the cross-section of the proposed structure; as depicted, the sensor comprises an analyte channel acting as the sensing medium, a plasmonic Gold (Au) layer, and cladding air holes of various dimensions. Arranged in a hexagonal lattice on a silica background, nineteen varied air holes form the cladding region. For this geometric setup, the radius of each hole was defined as $r_1 = 0.225\,\mu$m, $r_2 = 0.375\,\mu$m, $r_3 = 0.175\,\mu$m, and $r_c = 0.075\,\mu$m, while the pitch (Λ) remained fixed at $2\,\mu$m.

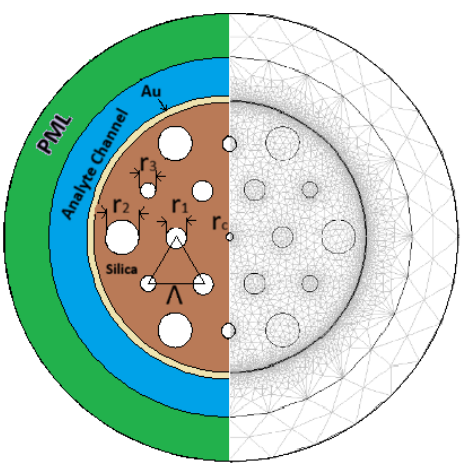

Figure 1. The cross-section of the proposed PCF based SPR sensor.

Light is effectively confined within the core region during the fundamental mode propagation. However, a specific fraction of light must leak toward the plasmonic layer to satisfy the conditions for resonance phase-matching, where the sensitivity measurement occurs. In this context, the light guidance is controlled by the various sized cladding holes, where each set of holes uniquely influences the sensor's propagation characteristics. The Gold (Au) material, serving as the plasmonic layer, was set to a thickness of 40 nm, with permittivity values sourced from the Johnson and Christy data. To determine the refractive index of the silica material, the Sellmeier Equation was applied [5]:

$$n^2(\lambda) = 1 + \frac{B_1\lambda^2}{\lambda^2 - C_1} + \frac{B_2\lambda^2}{\lambda^2 - C_2} + \frac{B_3\lambda^2}{\lambda^2 - C_3} \tag{1}$$

where λ represents the operating wavelength, n is the Refractive Index (RI), and $B_{1,2,3}$ and $C_{1,2,3}$ are the standard Sellmeier constants. Confinement losses reach their peak during the resonance phase-matching condition. Consequently, the precise calculation of these losses is vital, for which the following equation was utilized in this study [1]:

$$L_c = \frac{40\pi}{\ln(10)\lambda}\mathrm{Im}(n_{eff}) \times 10^4 \text{ [dB/cm]} \tag{2}$$

where $\mathrm{Im}(n_{eff})$ denotes the imaginary part of the effective RI.

III. Methodology

In this section, we detail the data preparation pipeline and the mathematical foundations of the regression models employed in this study. Given the high-fidelity nature of the FV-FEM simulations, the methodology focuses on maintaining physical consistency while maximizing predictive accuracy in a small-data regime.

A. Dataset and Preprocessing

The FV-FEM numerical analysis provided a labeled dataset containing 432 samples. The input feature space comprises six physical variables: Analyte Refractive Index ($n_{analyte}$), Operating Wavelength (λ), Pitch (Λ), and three distinct air hole diameters (d_1, d_2, d_3). Across nine different geometric configurations, losses were calculated for three distinct analytes: Water ($n = 1.33$), Ethanol ($n = 1.35$), and commercial hydrocarbon mixtures ($n = 1.36$).

To ensure rigorous comparison with existing literature, the model was evaluated using a static 7-1-1 split based on geometric configurations. The 432 samples were divided into 9 discrete geometric designs. Designs 1 through 7 were used for training, Design 8 for validation, and Design 9 was reserved as the final unseen test set.

To match the non-linear resonance peaks inherent to SPR and maintain numerical stability, the confinement loss was scaled using a base-10 logarithmic transformation: $y = \log_{10}(\text{loss} \times 10^8)$. All input features were scaled using Min-Max Scaling to strict $[0, 1]$ bounds to perfectly preserve the physical limits of the fiber geometry.

B. Model Architecture and Optimization

In this study, we evaluate two boundary-focused algorithms: Support Vector Regression (SVR) and Gaussian Process Regression (GPR). Unlike deep learning models, these algorithms are specifically designed to excel when the number of training samples is limited.

1) Support Vector Regression: SVR aims to find a function $f(x)$ that has at most ϵ deviation from the actually obtained targets y_i. The optimization problem is formulated as:

$$\min_{w,b,\xi,\xi^*} \frac{1}{2}||w||^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \tag{3}$$

subject to:

$$\begin{cases} y_i - \langle w, x_i \rangle - b \leq \epsilon + \xi_i \\ \langle w, x_i \rangle + b - y_i \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \tag{4}$$

where C is the box constraint and ξ_i, ξ_i^* are slack variables. We utilize the Radial Basis Function (RBF) kernel, $K(x_i, x_j) = \exp(-\gamma||x_i - x_j||^2)$, to map input features into a high-dimensional space for non-linear regression.

2) Gaussian Process Regression: GPR assumes function values follow a Gaussian Process prior: $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$, where $m(x)$ is the mean function and $k(x, x')$ is the covariance function. For a new input x_*, the predictive posterior mean $\bar{f}_*$ and covariance $\text{cov}(f_*)$ are:

$$\bar{f}_* = K_*^T[K + \sigma_n^2 I]^{-1} y \tag{5}$$

$$\text{cov}(f_*) = K_{**} - K_*^T[K + \sigma_n^2 I]^{-1} K_* \tag{6}$$

where K is the training covariance matrix and K_* is the train-test covariance. A WhiteKernel noise regularizer is added to the RBF kernel for stability.

IV. Results and Discussion

A. Predictive Performance: SVR vs. Deep Learning

The optimized SVR and GPR models demonstrated exceptional predictive capabilities under the exact testing methodology used in previous literature. As shown in Table I, our lightweight traditional models achieved superior accuracy without the need for complex data augmentation.

Table I. Benchmarking Proposed Models against Literature

Model	Avg. MSE
Baseline DNN (Zelaci et al. [1])	0.894
GAN-Augmented DNN (Zelaci et al. [1])	0.314
Proposed Optimized SVR	**0.121**
Proposed Optimized GPR	**0.185**

This performance gap highlights a fundamental difference in how these models handle small-data regimes. While Deep Neural Networks (DNNs) are highly complex global curve fitters, they often struggle when training samples are sparse, as is common with expensive FV-FEM simulations. In contrast, SVR hyperspecializes on the boundary conditions. By identifying the critical support vectors that define the transition between high and low confinement loss, SVR constructs a mathematically precise regression tube that generalizes better to unseen geometric designs than a global approximation.

B. Explainable AI (XAI) for PCF Design

SHAP analysis provided direct engineering insights that align with optical physics. As illustrated in Fig. 2, outer air hole diameters (d_2, d_3) are the primary drivers of confinement loss.

Physically, this makes sense because the outer cladding layers are responsible for suppressing the evanescent field leakage from the core to the plasmonic surface. Larger outer holes increase the refractive index contrast between the core and the cladding, thereby enhancing light confinement. Conversely, the inner hole (d_1) has a minimal impact on the overall loss gradient as it primarily influences the fundamental mode's effective area rather than the leakage rate toward the Gold layer. This "white-box" insight confirms that the SVR model has captured the underlying physical behavior of the PCF-SPR sensor.

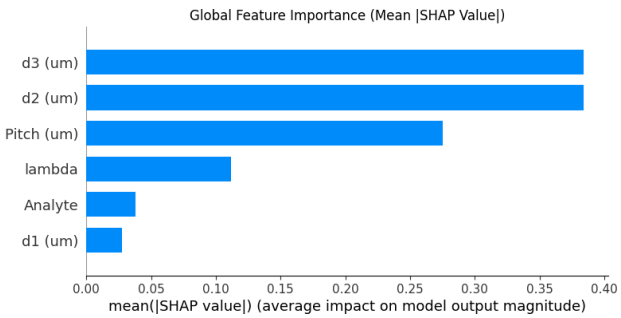

Figure 2. Global feature importance ranking derived from SHAP values.

V. Engineering Validation

To verify the physical consistency of the SVR model beyond simple error metrics, we performed an engineering validation by predicting the complete spectral loss response for varying analytes. As shown in Fig. 3, the model accurately captures the resonance peak shifts as the analyte refractive index (n_a) increases from 1.33 to 1.36.

The predicted average spectral sensitivity is 27.63 nm/RIU. While this value is relatively low compared to ultra-sensitive optimized designs in literature, it is highly consistent with the specific geometric parameters of the base design used for this validation fold. More importantly, the graph proves that the model correctly understands the phase-matching condition between the fiber mode and the surface plasmon polariton (SPP). As n_a increases, the resonance wavelength (λ_{res}) shifts toward longer wavelengths (redshift), which is a characteristic behavior of PCF-SPR sensors.

The model's ability to maintain these linear shifts across the entire spectral range, despite the highly non-linear nature of the individual loss curves, demonstrates that it has not merely "memorized" the dataset but has instead captured the fundamental physical relationships governing the sensor's operation. This confirms that the proposed SVR is a reliable surrogate model for rapid sensor optimization.

VI. Conclusion

This study proves that optimized SVR (MSE 0.121) and GPR (MSE 0.185) models provide a more accurate and sta-

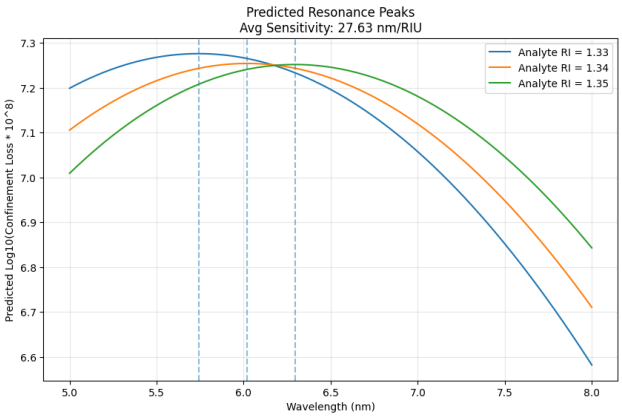

Figure 3. Predicted resonance peak shifts for varying analytes, demonstrating the model's ability to capture the physical phase-matching behavior.

ble alternative to deep learning for PCF-SPR sensor design. By strictly following the 7-1-1 configuration split used in literature, we demonstrate that traditional machine learning techniques can outperform GAN-augmented pipelines while requiring zero synthetic data generation.

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